UNICITY

SECURING THE AGENTIC ECONOMY.

Machines pay each other across cards, wallets, APIs and stablecoins — authority, permission and settlement evidence stay scattered. Unicity makes the proof travel with the payment.

A SEED PROPOSAL FOR GREG KIDD

AGENTIC PAYMENTS ARE NO LONGER A DEMO. THE MACHINES HAVE STARTED PAYING EACH OTHER DIRECTLY.

57.5%

OF WEB REQUESTS ARE AUTOMATED, NOT HUMAN — CLOUDFLARE, 2026

Payments are messaging on steroids — and the machines have started sending. About **100 million on-chain payments crossed Base in nine months** (Chainalysis); Mastercard now builds payment infrastructure for machines. The amounts are small. The frequency is high. The path may cross cards, wallets, APIs and stablecoins. The problem is trust.

> Who authorised it, who may receive it, the limit, why it settles — the proof rides with the money.

03 · THE STRUCTURAL LIMIT · THE SHARED LEDGER

A SHARED LEDGER WAS BUILT FOR HUMANS. MACHINES DON'T WORK THAT WAY.

Slow, sequential, contested — that is human finance, and the shared ledger is built for it. Proof-of-work, proof-of-stake, DPoS: each one must broadcast, order, validate and record every transaction, for everyone. When billions of sub-cent machine payments arrive, every node still processes every one.

BROADCAST

Every node hears every transaction.

ORDER

Global agreement on the sequence of all of them.

VALIDATE

Every node re-runs every transaction.

RECORD

Every node stores the whole global state, forever.

> The bottleneck is the design, not the implementation. So Unicity removes the shared ledger.

FAIR ACCESS TAKES THREE THINGS. THE THIRD WAS NEVER BUILT.

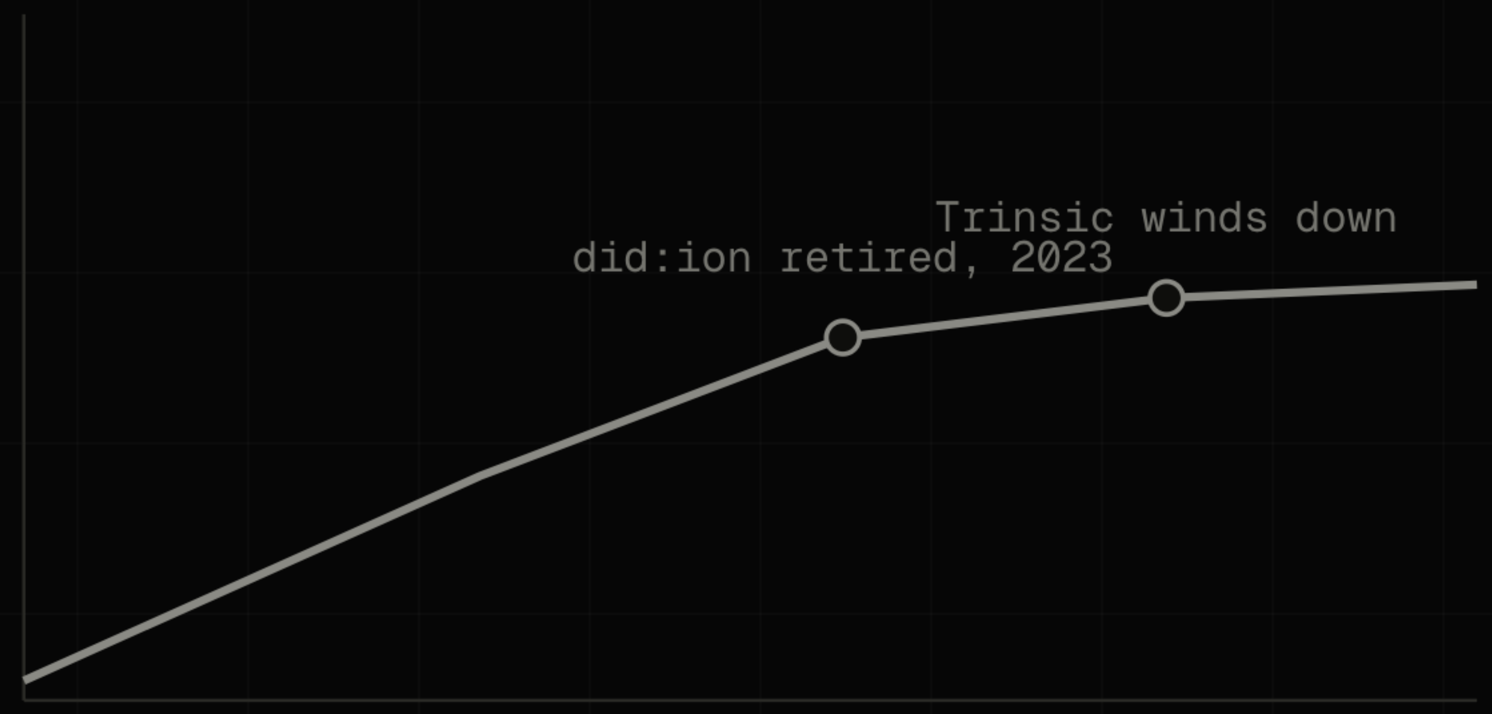
01	**CUT OUT THE MIDDLEMAN** A decade of open infrastructure settled it.	SOLVED
02	**LET VALUE MOVE FREELY** Stablecoins now settle tens of trillions a year.	SOLVED
03	**PROVE WHO IS ON THE OTHER END** The payment itself knows who may receive it.	STILL OPEN

The third is the problem that decides fair access. It is the one Unicity builds.

IDENTITY REACHED THE RIGHT ARCHITECTURE BEFORE THE MARKET COULD CARRY IT.

Self-sovereign identity had the third problem right, early: identity is permission to act. The field stalled at the same wall — did:ion retired in 2023, Trinsic wound down. The credential sat beside the transaction, and a check that can be skipped gets skipped.

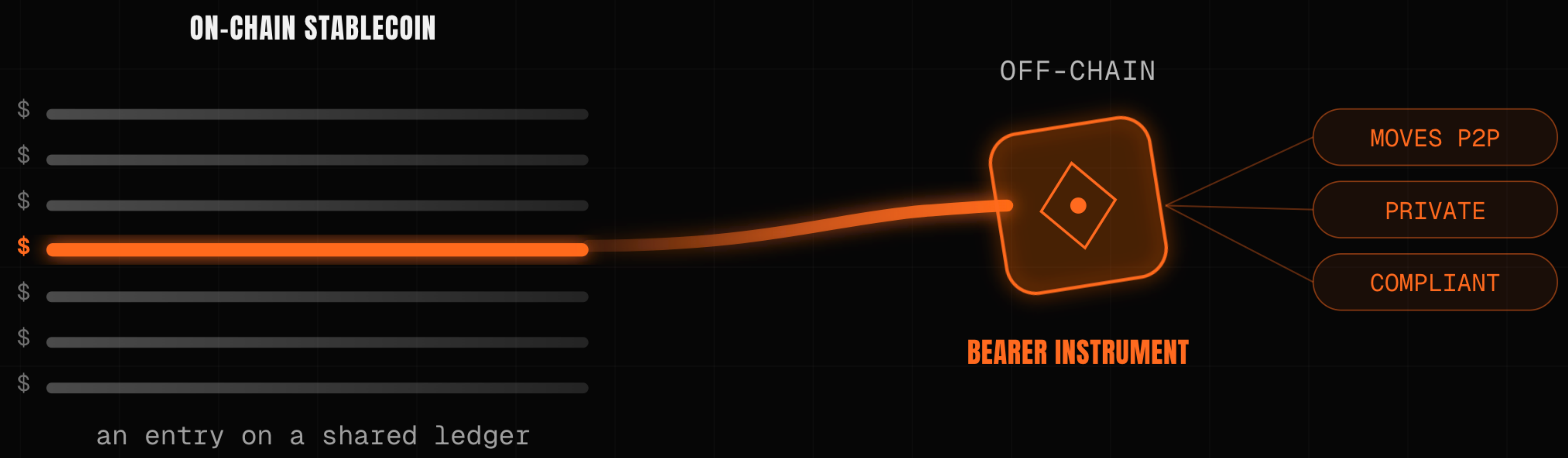
USBC drew the right conclusion: put identity inside the dollar. On one charter it settles. Between charters — between strangers — it cannot. Closing that gap is the build.



digital identity — climbs, then flattens

A DOLLAR ON-CHAIN IS A ROW IN A LEDGER SOMEONE ELSE KEEPS.

Unicity makes it cash. The dollar becomes a bearer object — a native data type the system holds, like a file. It carries its own value and its own proof, moves directly between two parties, and verifies on arrival against the token alone. No chain to call.



SELF-CONTAINED

The value lives in the object, not a row a network keeps for it.

SELF-PROVING

It carries its own proof of validity, minted against a locked source-chain asset.

VERIFIES ON ARRIVAL

The recipient checks it locally, offline, over any channel, with no chain to call.

07 · HOW IT WORKS · ONE QUESTION

THE ONLY JOB A CHAIN EVER HAD: HAS THIS BEEN SPENT?

Take everything else away and one job remains — stop the same money being spent twice. The network receives an opaque commitment and answers exactly that: spent, or unspent. Never the data, the parties, or the amount. Hard cryptography — one trivial question.

CONSENSUS & EMISSION

decentralized base security

↓ certified state root

UNIQUENESS ORACLE

stores only proofs that token states are unique

↓ inclusion proofs

EXECUTION LAYER

agents and users transact client-side, off-chain

OPAQUE COMMITMENT

The oracle sees only spent or unspent, never the trade itself.

PROVES THE TREE, NOT THE TRADE

A zero-knowledge proof attests the sparse-Merkle-tree state transition, not each payment.

THREE PROOF TYPES

Inclusion (registered, eternal), exclusion (low-latency settlement), unicity (the tree has not forked).

EVERYTHING ELSE MOVES TO THE EDGE. ONLY UNIQUENESS STAYS.

Everything a normal blockchain does for everyone — communication, storage, validation — moves to the parties who actually care, handled with ordinary tools. Each transaction is verified on its own, independent of every other one. That relocation, not a throughput number, is why it scales.

COMMUNICATION

The asset carries its own proof; the wire just moves bytes — NOSTR, email, QR.

STORAGE

State lives with the owner, not on every node — browser, device, IPFS.

VALIDATION

The recipient verifies; no validator set in the path — client-side, in parallel.

A named primitive, not a novelty: client-side validation and single-use seals (Bitcoin Optech, RGB, Nervos CKB).

THE UNBUNDLING OF THE BLOCKCHAIN. WE TOOK IT TO THE END.

Each generation of consensus removed work from the network. This is the next step, not a departure.

2009 · BITCOIN

CORRECTNESS + ORDERING

Every node certifies every transaction and agrees on the order.

2023 · SUI · FASTPAY

CORRECTNESS ONLY

Global ordering removed for assets that share no state.

2026 · UNICITY

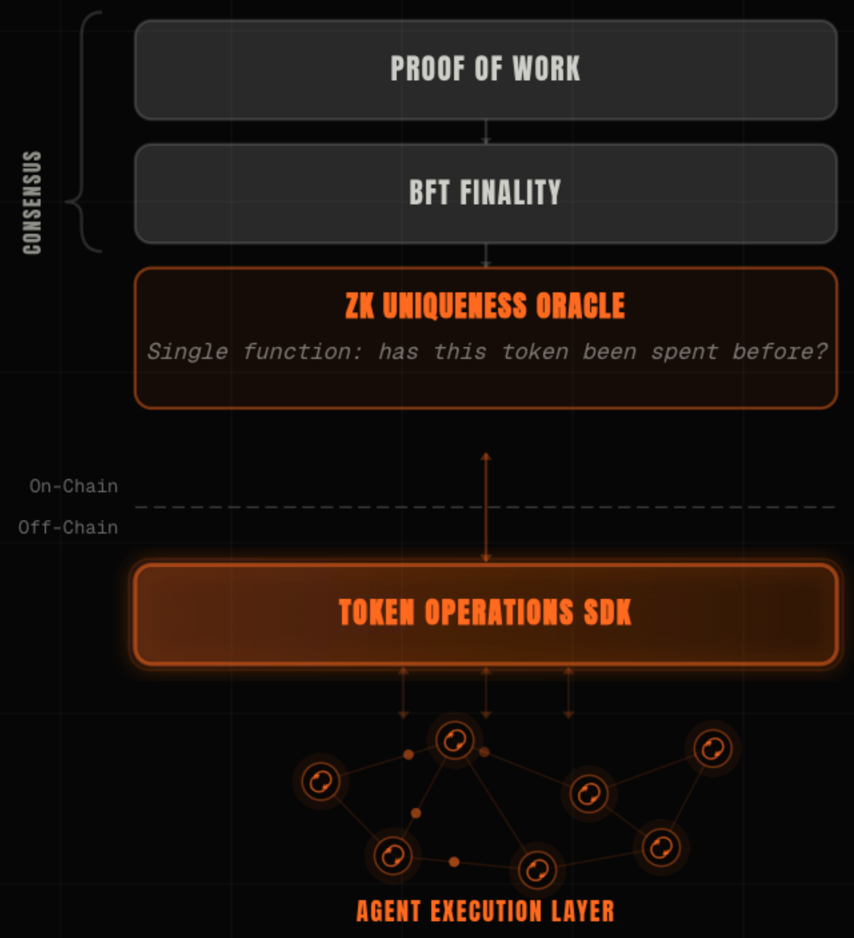
UNIQUENESS ONLY

The network attests one thing — has this been spent. Correctness moves to the edge.

> The logical endpoint of a decade of consensus research — not a competitor to chains.

A MINIMAL CHAIN, AND AN ECONOMY THAT RUNS OFF IT.

The consensus layer stays deliberately small. Everything a user touches is off-chain, joined by a thin SDK. The chain uses proof-of-work for one reason — to copy Bitcoin's trust model: trust no one.



ON-CHAIN, MINIMAL

RandomX proof-of-work, then chained HotStuff for ~1s finality, then the uniqueness oracle.

THE SDK

A thin bridge in JS, Java and Rust between off-chain agents and on-chain proofs.

AGENT EXECUTION LAYER

Transport (I2P, TCP/IP), messaging (NOSTR), storage, runtime — all peer-to-peer.

THE RULE LIVES INSIDE THE MONEY.

Each token carries its own receive-condition — who may receive it, the jurisdiction, the limit. A transfer that fails it cannot be constructed. The asset attests its own compliance, never the private data behind it. The proof travels with the payment.

PROGRAMMABLE

Signature, multi-sig, k-of-n threshold or time-lock, written once and travelling with the token.

PROTOCOL-ENFORCED

The check lives in the asset, not an app or custodian — no intermediary to bypass.

RISK MANAGED, NOT AVOIDED

The issuer sets who may hold the token; a rogue counterparty is on the platform and contained, not kept out.

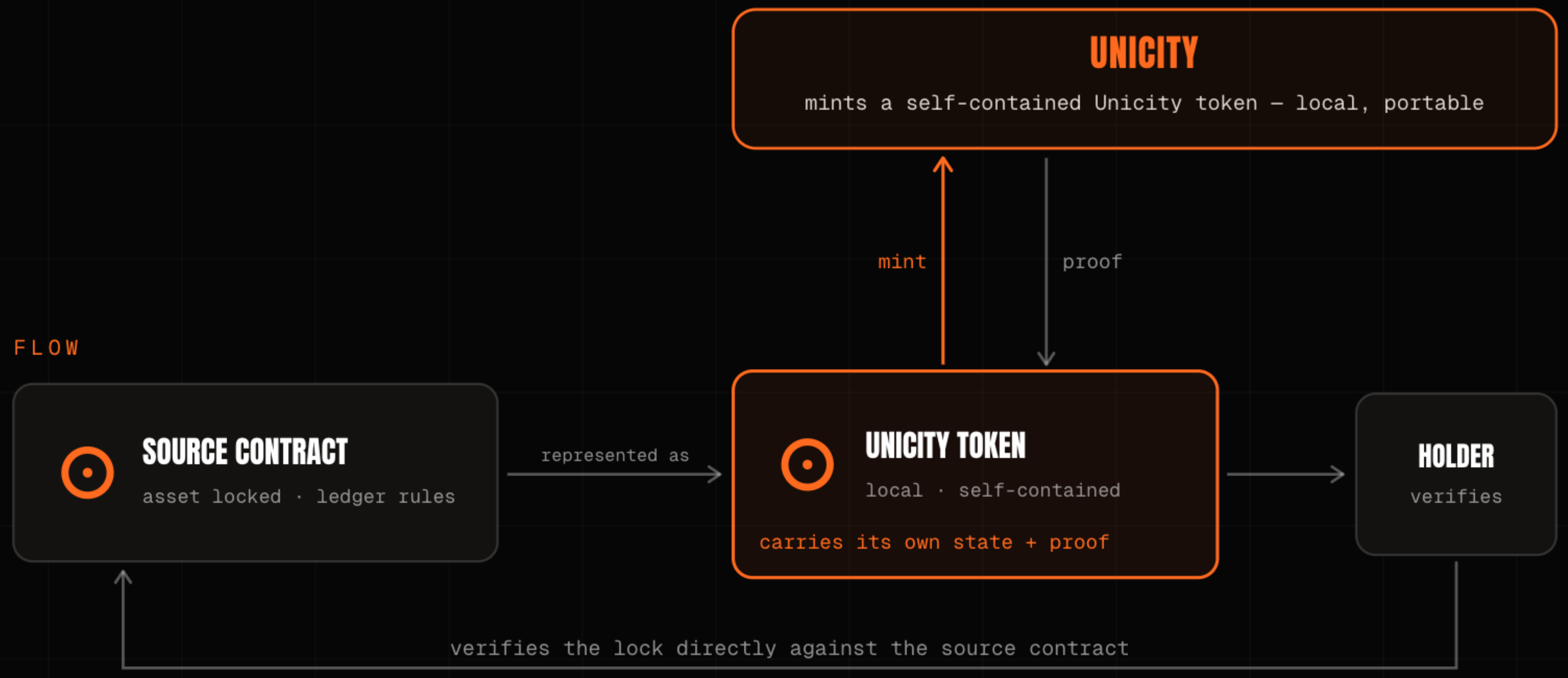
TOTAL PRIVACY BY DEFAULT – AGAINST THREE OBSERVERS, PROVEN.

Privacy was a founding principle of the design, not a setting added later. The papers prove it — against three concrete adversaries.

THE NETWORK	Sees only opaque, perfectly-hiding commitments — never amounts, parties, or balances — and cannot link two consecutive transactions of the same token.	PROVEN
THE SENDER	Cannot watch when, or to whom, the recipient later spends what was sent.	PROVEN
ANYONE WITH THE ADDRESS	One published key; every sender derives a fresh, indistinguishable transaction key against it.	PROVEN

NO BRIDGE. NOTHING TO HACK.

Bridge hacks are the largest loss category in crypto — a bridge holds the asset and forwards a message, and a message can be forged. Unicity holds nothing. A locked source-chain asset becomes a self-contained token, and the recipient verifies the lock directly against the source contract.



NO BRIDGE
There is no cross-chain message to forge and no pooled liquidity to drain.

NO CUSTODIAN
No third party holds the asset between chains; the holder verifies the lock itself.

NOTHING TO HACK
The most dangerous middleman in crypto is structurally gone.

FIVE WINS.
ONE HARD PROBLEM.

Remove the shared ledger and five things follow — speed, scale, privacy, compliance, no bridge. One remains. With no shared state, the tokens themselves must enforce atomicity — and that is the trade.

THE HARD PART OF CASH-LIKE MONEY IS THE TRADE.

Two parties swap, with no one in the middle to hold both legs. The old fix is a timed lock — and the clock is the attack surface, because one side can stall and walk. Unicity drops the clock: both commit independently, and the swap forms for both at once, or not at all.

HASHED TIMELOCK (HTLC)

1 · A locks, reveals hash y

2 · B locks against same y

3 · A claims, revealing preimage x

4 · B must claim before timeout

THE TIMING TRAP

B late on step 4 — a dropped connection — and A keeps both.

UNICITY PREDICATE SWAP

A COMMITS
locks their token

B COMMITS
locks their token

UNICITY SERVICE
the shared reference

AUTOMATIC FAIRNESS

✓ both locked → swap completes
✓ either walked away → both refunded
✓ no deadlines, no chasing, done

NO CLOCK

Both commit independently against a shared reference: no deadline, no refund timer.

BOTH OR NEITHER

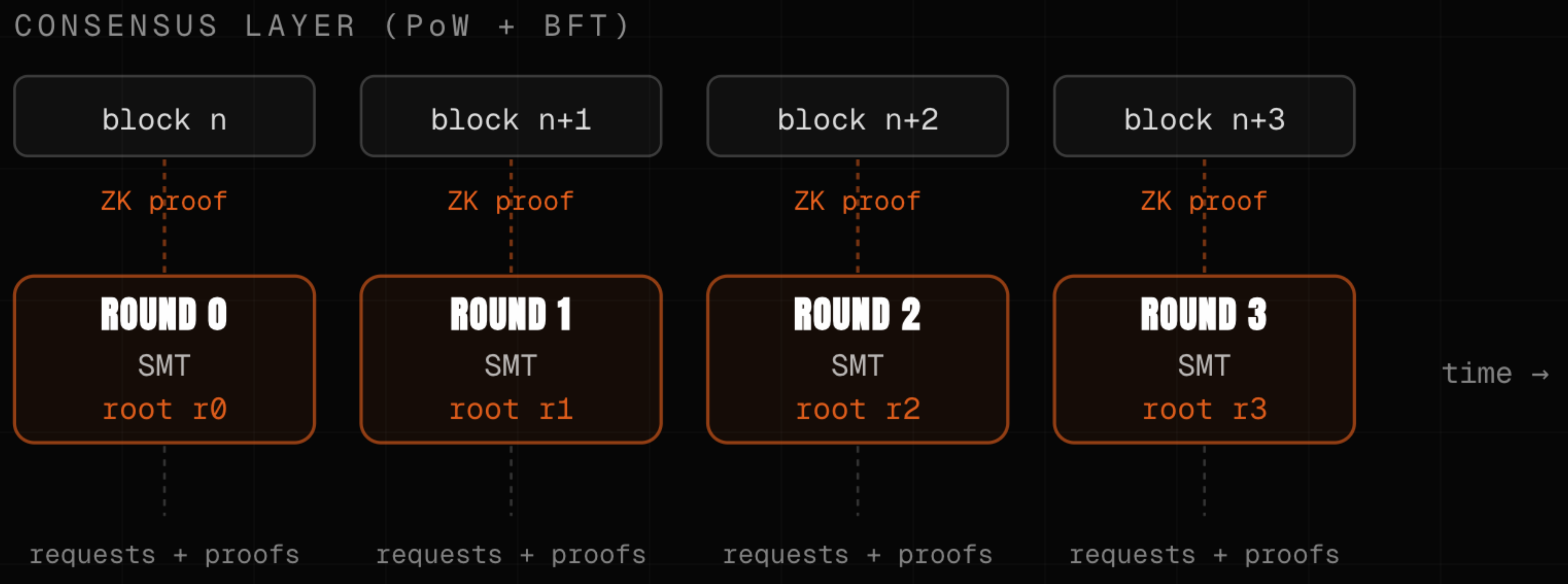
Both locked, the swap completes; either walks away, both keep their token. Capital is never trapped.

NO MEMPOOL, NO MEV

It runs off-chain at machine speed, immune to front-running and mempool games.

THE ORACLE VALIDATES NOTHING. THROUGHPUT IS ADDED, NOT RATIONED.

The oracle re-executes nothing — it certifies one consistency proof per round, not each payment. Cost per transaction collapses. Capacity grows the way servers do: add a shard. Permissionless and redundant, so it is not a new chokepoint.



ADD A SHARD, +30,000 TX/SEC

A design figure on a single consumer CPU — Plonky3 AIR, no trusted setup. Stated, not a live benchmark.

SUB-MICROCENT PER PROOF

Each new shard adds capacity; the proofs stay succinct.

PERMISSIONLESS, REDUNDANT

Independent staked aggregators join freely; sub-trees run in parallel, with no global bottleneck.

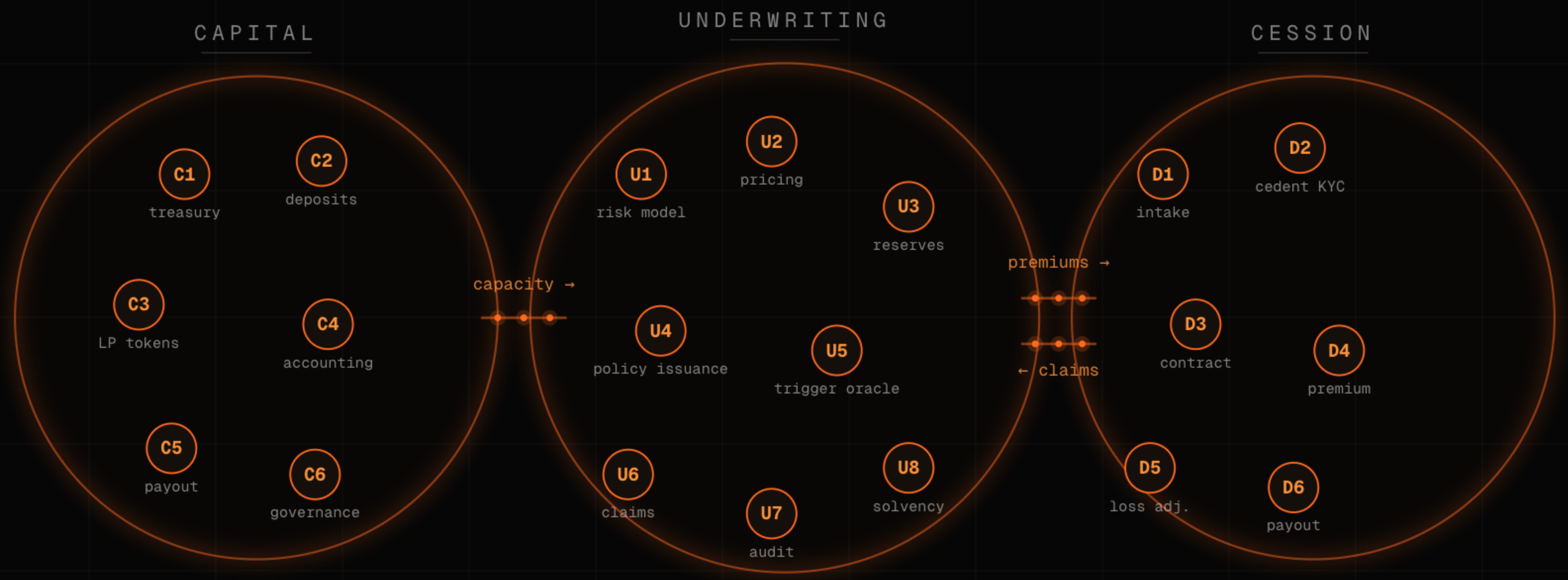
CEX SPEED. DEX CUSTODY. PRIVATE AND COMPLIANT.

When two agents settle directly, the venue has nothing left to do. Binance has the speed. Uniswap has the custody. Neither has the privacy, and neither was built for a machine payment. Unicity has all of it, with no venue at all.

	BINANCE	UNISWAP	HYPERLIQUID	UNICITY
SPEED	sub-second	block time	fast	sub-second
SELF-CUSTODY	custodial	you hold keys	you hold keys	you hold keys
PRIVACY	operator sees all	fully public	public positions	unlinkable
MEV	operator order	sandwiched	mempool	no mempool
COMPLIANCE	off-chain KYC	none	none	in the asset

A WHOLE COMPANY, RUN BY A SWARM OF AGENTS.

Not a roadmap — we built one. A decentralized autonomous corporation for BlackRock: a weather-based parametric insurer whose capital, underwriting and cession all run by agents transacting in Unicity tokens. Agents are the new smart contracts — verifiable code, off-chain, on bearer assets.



CAPITAL

Capacity is committed and tracked on-chain-proven bearer tokens.

UNDERWRITING

Premiums priced and bound by predicate to the policy's receive-condition.

CESSION

Claims and reinsurance settle agent-to-agent, with no venue in the middle.

THE SERIOUS PLAYERS ARE SPENDING BILLIONS TO ESCAPE THE LEDGERS THEY BUILT.

AP2 authorises, x402 settles, the players below optimise the ledger — the mandate and the audit trail scatter across layers. They split the proof. Unicity keeps it in the asset.

PLAYER	THEIR MOVE	APPROACH
CIRCLE	Pivoted to infrastructure — the Arc L1, a $222M presale, bought Malachite.	optimise the ledger
SOLANA	Alpenglow cut finality to ~150ms — and its own co-founder concedes a physical ceiling.	optimise the ledger
CARDANO	Midnight adds zero-knowledge selective-disclosure privacy.	optimise the ledger
USBC · GREG KIDD	Put identity inside the dollar — the right conclusion, on one charter.	the right idea
UNICITY	Removes the shared ledger at the data-structure level.	eliminate it

WE BUILT THIS ONCE BEFORE. NOW MAKE IT HOLD BETWEEN STRANGERS.

THE TEAM

The cryptographers behind Unicity built Guardtime / KSI — sovereign-grade verification deployed with the Estonian Government, Lockheed Martin, Boeing and NATO. In production since 2012, eIDAS-grade. KSI held **300,000 tx/sec** in Eesti Pank's 2021 test — the lineage, not a live Unicity number.

THE PROOF

Three math papers prove what matters — privacy and no-double-spend. Drop them in any model and they check. Throughput is a sharding design result, named as such. All public on github.com/unicitynetwork.

THE ASK

$5M Seed · $25M cap / $50M token FDV · SAFE + token warrant. First build: a USBC dollar that settles the moment the credential checks out — Unicity brings the settlement; the partner brings the regulatory layer we openly lack.

ONE CHARTER SETTLES AT HOME. THIS IS THE LAYER THAT CARRIES THE RULE AND THE PROOF BETWEEN TWO BANKS — OR TWO MACHINES — THAT SHARE NOTHING ELSE.